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Policy instrument designed to gain transition legitimacy: A case of Chinese nuclear development

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ABSTRACT

Nuclear energy development may encounter significant opposition if it is not deemed as legitimate by the general public even in an authoritarian state like China. This paper collects policy instruments designed in four nuclear stations in China, and reports both advantages as well as disadvantages of a centralised authority. At the central level, four types of policy tools are designed to win nuclear legitimacy. At local level, however, education and monitoring tools are well implemented with the help of highly centralised administrative system, while information disclosure and participation tools are only partially implemented since they need innovative reform of citizen deliberation and involvement mechanism. Instead, local government and nuclear firms design various legitimacy-oriented policies such as institutionalizing nuclear into local culture and social life. However, there still lacks of effective policy design for long-term legitimacy winning in China that clearly defines the concept of "the public" and "legitimacy".

1. Introduction

China is currently transitioning its coal-based energy structure to a non-fossil fuel-based low-carbon energy structure (i.e., using nuclear power, wind and solar photovoltaic cells). In its Intended Nationally Determined Contribution (INDC), China committed to achieving peak carbon dioxide emissions by approximately 2030, or sooner as its best efforts allow, which implies a reduction in carbon dioxide emissions by 60% to 65% per unit of GDP from 2005 levels. To reach this goal, China plans to increase the share of non-fossil fuels in its primary energy mix to approximately 20% by 2030.[1] Nuclear energy is an important source of low-carbon energies China plans to develop.[2] Like other countries facing sustainability transitions, China faces challenges not only in technological innovation, but also in making social and political changes (Avelino et al., 2016).

On the one hand, China has proposed an aggressive nuclear development plan, aiming to achieve a nuclear power capacity of 58 GW in service and a capacity of 30 GW under construction by 2020 (State Council, 2012). Given that China had an operational capacity of only 33.5 GW and a capacity of 23.4 GW under construction at the end of 2016, this aggressive long-term plan implies a fast construction speed: more than 10 power stations with an installed capacity of 1 GW must be built every year for the next 4 years, and a number of inland nuclear power plants must be built, although they are still technically under debate (Liu et al., 2017; Wen

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[1] Source: UNFCCC INDC document webpage, http://www4.unfccc.int/Submissions/INDC/Published%20Documents/China/1/China's%20INDC%20-%20on%2030%20June%202015.pdf. Retrieved on 30 July 2017.

[2] Even though China has made tremendous progress in developing solar and wind energy, it is unlikely that the country will meet its INDC goal purely with these renewable forms of energy. Nuclear energy is stable and could be constructed near its end users. Therefore, nuclear energy is inevitably emerging as one important part of the sustainability transition solution.

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2014).[3]

Meanwhile, public opposition to the siting and operation of nuclear stations has increased in recent years. First, Chinese citizens are becoming concerned about the potential risks to public health, safety and the environment posed by nuclear projects. Although there is no systematic record of Chinese public opposition to nuclear stations, scattered research shows a relatively high level of nuclear acceptance in China before the Fukushima accident (Liu et al., 2008; Zhou and Zhang, 2010) and an observable slump right after the accident (Huang et al., 2013). Sun et al. (2016) used survey data to measure uncertainty about nuclear technology among the general public and declared that fewer people considered nuclear power to be safe in China. What is more, the technological trajectory of inland nuclear technology, which has proved to be as safe as coastal nuclear technology in many other countries, has raised uneasiness among the general public and brought challenges to policy design (Zhou et al., 2011; Chen 2010).

Second, the general public has become increasingly involved in the nuclear policymaking and implementation process. Although the International Atomic Energy Agency (IAEA) has made much effort to involve the public in nuclear power decisions,[4] the Chinese public had no direct opportunity to participate in nuclear decisions before 2014, due to China's highly centralised political system. With the rapid development of Internet technology, the general public can now express opposition directly via online commenting, make calls to government officials, protest and even riot. For example, more than a thousand people protested the construction of a nuclear fuel production plant in Jiangmen, Guangdong Province in 2013, which led directly to the cancellation of the project (Sun and Zhu 2014). Local tensions over nuclear safety in Pengze (Jiangxi Province) and Cangnan (Zhejiang Province) also ran high, with the result that both projects were postponed (King and Ramana 2015).

The topic of transition legitimacy is normally discussed in the context of Western liberal democracies. Studies have reported that the general public may not accept sustainability transition visions and policies (Hendriks 2009; Avelino et al., 2016). Researchers have attributed this partly to the complexity and long-term nature of transition tasks and partly to the limitations of concurrent democratic mechanisms (Hendriks 2008, 2009). Confusion in the key concepts and basic underlying assumptions of democracy have led some researchers to call for the redefinition and reframing of democracy, while others consider a centralised decision-making or exclusive political process suitable for sustainability transition goals (Hendriks 2009; Heilbroner 1974; Jhagroe and Loorbach 2015). However, non Western liberal democratic contexts like China, have been under-researched in transitions studies in terms of issues around participation and legitimacy.

This paper reports on the Chinese experience to fill this gap in the literature. To ease the tension produced by public opposition, the Chinese government has to adjust its exclusive decision-making system to some extent and make the general public an important stakeholder in the nuclear policy process (He et al., 2013). Meanwhile, it relies heavily on a top-down decision-making pattern as a means of gaining legitimacy as it designs and implements policy instruments. Can the inclusive effort and exclusive system work in harmony? Can Western theories of transition legitimacy help to solve Chinese problems? Have any non-Western instruments been designed to promote legitimacy? If so, how could they influence nuclear legitimacy in China?

This research examines policy instrument designs in four power plants in different locations, socio-economic situations, ownership situations and stages of development. The rest of the paper is organised as follows. Section two introduces the legitimacy challenges that Chinese nuclear energy faces, and also summarises the key policy tools designed to gain nuclear legitimacy along two dimensions: education-oriented tools and political trust-oriented tools. Section three discusses four cases in China, covering all of the legitimacy tools. Detailed case information is collected in each case and the main findings and discussion are reported in section four. Section five concludes the paper and raises further research questions about legitimacy in sustainability transitions.

2. Literature base and theoretical framework

2.1. Legitimacy challenges in sustainability transitions

The road to sustainability poses a significant challenge to human society, and particularly to governance and policy systems. Because the concurrent institutions and policies are mainly designed for short- and middle-term goals, vast uncertainties appear when we try to reach the long-term and ambiguous goal of sustainability, which requires fundamental changes in governance routines and social values (Loorbach, 2007, p. 11). The new study area of sustainability transitions has emerged to allow researchers to understand, interpret, analyse and intervene in complex social systems in an effort to smooth transitions (Loorbach 2007, 2010; Hendriks 2009; Avelino et al., 2016; Jhagroe and Loorbach 2015). One argument in sustainable transition research is that the political legitimacy of transition solutions might be undermined by traditional democratic assumptions. Hendriks (2008, 2009) studied how practitioners ensure that sustainable policy procedures and outcomes are accountable to and legitimated by the public in the Dutch transition process and found that traditional democratic assumptions had been taken for granted and had been transferred to transition management without verification. For example, it was assumed that democracy, no matter which type,[5] provided a procedural context through which social participants could express their preferences on sustainability and push for legitimate decision-making. If this assumption were true, policies' legitimacy would be automatically ensured. As an extension of this assumption, sustainable policies would still be legitimate even if decisions were made by energy experts and elites with limited or no opportunities

[3] A detailed distribution of concurrent nuclear plants is available upon request.

[4] Source: IAEA news at https://www.iaea.org/NuclearPower/News/2013/2013-09-10-inig.html. Retrieved on 2 August 2017.

[5] Various formats of democracy have been discussed within the sustainable transition area, including representative democracy, deliberative democracy, participatory democracy, post-foundational democracy and others (Hendriks, 2008, Hendriks, 2009; Jhagroe and Loorbach 2014; Avelino et al., 2016). This paper does not differentiate between these formats.

for direct participation (Hendriks 2009).

In practice, sustainability transition practices might not follow these assumptions for various reasons. Hendriks and Grin (2007) studied the Dutch Energy Transition Program and found that the meaning of legitimacy was neither clear nor stable within the democratic system. Rather, it was constantly discussed and negotiated. In a later study, Hendriks (2009) found that the concept of legitimacy was defined differently when used in different democratic theories, such as technocracy and elite theory, representative democracy and interest group pluralism. Chilvers and Longhurst (2016) enhanced this observation by checking UK low-carbon transition programs; they found that many elements in democratic participation, such as participation exclusion, output legitimacy and even the definition of the 'public', were orchestrated and mediated.

These studies called for the deconstruction and reframing of traditional democratic notions and democratic practice. Jhagroe and Loorbach (2015) argued that some inherent features of the transition process are not traditionally democratic. For example, transition practices are likely to be selective and exclusive, given the assumption that large-scale structural changes are inherently in conflict with existing interests and powers. Democracy, an ideology that has guided Western countries for a long time, needs to be adjusted for sustainability transitions. Yet no neat solutions have been presented in the literature (Hendriks 2009; Chilvers and Longhurst 2016; Dryzek 2001).

Sustainability transitions in China present numerous theoretical challenges. As a centralised authoritarian state, similar to the strong central authority proposed by some green theorists in response to the ecological crisis of the 1970s (e.g., Heilbroner 1974), China may be capable of dealing with long-term policymaking and sustainability transitions. However, China also faces its own legitimacy challenges in its transition period.

2.2. Legitimacy challenges in chinese nuclear energy development

China faces three legitimacy challenges in its nuclear energy transition process. The first one comes from the increasing public concern about the environment. Along with rapid economic development have come stronger environmental concerns among the public and a preference for balancing economic development and environmental protection (Economy, 2007). Citizens feel uncertainty about nuclear technology's impact on environment quality and human health and about the potential for accidents. Nuclear energy's legitimacy, therefore, has declined over the past two decades, especially since the Fukushima accident (Huang et al., 2013; He et al., 2013). Citizens' opposition has sometimes been expressed harshly. The Chinese National Nuclear Corporation (CNNC), in cooperation with a French company, Areva, tried to find a location for a nuclear fuel processing and recycling plant. Lianyungang, a city in north Jiangsu Province, was one of the options. Although the investigation was still in process, large-scale demonstrations broke out when local residents learned about it in August 2016. The investigation was cancelled and the Lianyungang government declared that the project would be indefinitely postponed.[6] Public opposition also led to protests in Jiangmen (Guangdong Province), Pengze (Jiangxi Province), Rushan (Shandong Province) and other places (He et al., 2013).

Second, the Chinese nuclear industry experienced an identity conversion that also led to a legitimacy crisis. The first Chinese nuclear power plant was launched as a military project in Qinshan, Zhejiang Province, in the 1980s. Because the decision-making process was entirely top-down, residents of the area near the Qinshan nuclear station had no chance to learn the details of the project and were not involved in the decision process. Legitimacy, however, was naturally guaranteed under the national security umbrella. Later, state-owned companies such as the CNNC, the Chinese General Nuclear Power Corporation (CGNPC) and the State Power Investment Corporation (SPIC) became the main entities developing Chinese nuclear energy. Without the veil of national security, Chinese nuclear energy has to win its legitimacy by following other standards such as participation and transparency (He et al. 2013).

Third, China is undergoing a liberalisation process with declining political trust. Unlike Western democratic societies, China has a centralised political and administrative system in which citizens have traditionally had a high level of political trust even though they have had little chance to participate (Li 2004; Cui et al., 2015). However, recent years have seen a trend of decreasing political trust as citizens value liberal democracy more and seek deliberation and citizen empowerment (Wang and You 2016). As a result, citizens are challenging the legitimacy of government decisions more than ever. Nuclear decisions are no exception.

With legitimacy challenges coming at individual level, industrial level, and society level, Chinese nuclear development calls for new policy design for its smooth development.

2.3. Policy instruments designed for nuclear legitimacy

Multiple public instruments have been designed to win the general public's acceptance of nuclear power in many countries (Goodfellow et al., 2011; Lee and Gloaguen 2015; He et al., 2013; Huang et al., 2013). These policy instruments are designed in two dimensions: nuclear education tools (Wildavsky and Dake 1990; Sjöberg and Drottz-Sjöberg 1991; Wang et al., 2013; He et al., 2014; He et al., 2013) and tools to increase political trust (Earle and Cvetkovich 1995; Siegrist and Cvetkovich 2000). Both of these dimensions represent essential elements in determining perceptions of nuclear energy, based on Western empirical studies.

2.3.1. Education-oriented legitimacy instrument design

There is no academic consensus on how knowledge influences rational decisions with regard to scientific technology. One school, following the technocracy and elite theory, holds that technological experts and elites are able to make rational decisions because

[6] Media reports can be found at https://www.ft.com/content/dacb775a-5c7f-11e6-bb77-a121aa8abd95 and https://www.nytimes.com/2016/08/09/world/asia/china-nuclear-waste-protest-lianyungang.html.

they are more knowledgeable than others, particularly on complex and sequential policy issues such as nuclear energy (Kuklinski et al., 1982; Wildavsky and Dake 1990). This evidence shows that people with more knowledge about nuclear energy are more likely to deem it legitimate and show higher levels of acceptance (Kuklinski et al., 1982; Sjöberg and Drottz-Sjöberg 1991; Wang et al., 2013; He et al., 2014; He et al., 2013). Meanwhile, others question the cause-and-effect pattern drawn here and argue that knowledge levels relate to legitimacy in an inverted U shape (Visschers and Wallquist 2013). In other words, the nuclear acceptance level increases when citizens obtain more nuclear knowledge, but drops when they learn still more about it. There is also disagreement on the role of scientific knowledge in policy decisions. Wynne (2010) argues that '(science)…has become the author of public meaning rather than the informant of public policy'. Therefore, he argues that negative feelings about nuclear energy reflect 'unanticipated consequences' and should be treated as a trust issue rather than a scientific issue.

Multiple educational tools were introduced at an early stage of nuclear development. Legitimacy-winning tools, including media discourse, school educational programs, nuclear technology exhibitions and contests, community training programs, site visits to nuclear stations and others, are still functioning in many countries (Han et al., 2015; Gamson and Modigliani 1989; Barbour and Gill 2014; Nakamura 2016).

The effect of technology education, however, might be influenced by many factors. Risk perception, which refers to an individual's capacity to perceive risk, normally depends on individual characteristics such as risk sensitivity, trade-offs among different risks and specific fears (Starr 1969; Sjöberg 2000; Bickerstaff et al., 2008). Not all of these individual features can be influenced by outside policy tools. Therefore, knowledge tools can only have limited effects on legitimacy.

2.3.2. Political trust-oriented legitimacy instrument design

From the point of view of democratic theory, the general public does not have to gain enough knowledge to make individual judgments about nuclear legitimacy (Wynne 1996). Rather, the public depends on other people or mechanisms to make decisions, such as technocrats and elites (following technocracy and elite theory), group discussions and group decisions (following deliberative democracy and participatory democracy theory) or the government and relevant interest groups (following representative democracy theory or interest group pluralism).

Chinese citizens rely on various entities to make their nuclear legitimacy decisions. Fang (2014), for example, studied why local villagers regard nuclear station plans as legitimate from a social-psychological point of view and found that uneducated villagers or uninformed people took government decisions and the conclusions of technical experts as substitutes for their own knowledge when they trusted the decision-makers. This trust may have come from the assumption that 'strong respect for status and authority is linked with the conviction that these authorities (*Chinese governments*) will do their best to reduce the risks to society to a minimum' (Renn and Rohrmann 2000). Fang (2014) quoted a villager as saying: 'we don't worry about that [*nuclear accident such as Chernobyl*]. The technology used by our country is more advanced. Chernobyl [*the accident*] happened in the Soviet Union. Their technology was too old. The technology used by our country is advanced',[7] which verifies this assumption. He et al. (2013) also observed a high level of political trust in the government with regard to nuclear information provision, emergency response to nuclear accidents and decision-making about the country's nuclear future. However, they attributed this trust to the strong information-control capacity of the authoritarian state.

Besides political trust, certain interests or value preferences regarding green development and climate mitigation also increase nuclear legitimacy among the public. When citizens conceptually connect nuclear development to popular values such as green electricity generation, technological innovation or climate mitigation, it is highly likely that they will accept the energy format without increasing their knowledge level or political trust (Bickerstaff et al., 2008; Wang et al., 2013; Sun et al., 2016).

Political trust and value preference-related policy instruments are designed accordingly: public hearings, conflict mediation, public opinion surveys, focus groups, citizens' juries, NGO collaborations, community involvement, collaborative governance and consensus conferences are typical instruments (Rowe and Frewer 2000; Adams and Hess 2001; Krütli et al., 2010).

3. Sample selection and methodology

This paper, then, tries to cover all of the instruments that the Chinese government and nuclear firms use to gain legitimacy for Chinese nuclear energy development in order to check if they design these instruments following the two dimensions discussed above. Data collection, then, was designed in two parts. First, legitimacy policies designed at the central level were collected and analysed. These policies provide strategic guidance to local governments and nuclear firms as they design detailed policy tools.

Firm-level data were then collected. CNNC, CGNPC and SPIC are the only three nuclear firms in China. Each firm builds nuclear power plants independently or jointly with other firms. Therefore, each nuclear power plant takes charge of public legitimacy communication. This paper thus uses the nuclear power plant as the unit of analysis and collects the legitimacy instruments designed by each plant as raw data.

Four nuclear power plants were selected: the Qinshan nuclear power plant in Zhejiang Province, the Daya Bay nuclear power plant in Guangdong Province, the Taohuajiang nuclear power plant in Hunan Province and the Pengze nuclear power plant in Jiangxi Province. These four plants are varied in terms of operational stage (two are awaiting construction,[8] while the other two have safely operated for more than 20 years), location (two are located in coastal regions, while the other two are inland nuclear plants),

[7] Brackets and italics added by the author.

[8] The construction of the Taohuajiang nuclear power plant and the Pengze nuclear power plant is currently pending due to the strategy set by the central government after the Fukushima accident. When the government restarted nuclear development in 2015, only coastal projects were approved.

Table 1
Case selection descriptions.
*Source: information collected online and in interviews by the author.

	Qinshan	Daya Bay	Taohuajiang	Pengze
Location (Province)	Zhejiang	Guangdong	Hunan	Jiangxi
Major owner	CNNC	CGNPC	CNNC	SPIC
Year of initial construction	1984	1987	2008	2009
Year of commercial operation	1991	1994	Pending	Pending
Local city GDP in 2015 (billion RMB)	351.7	1750	135.4	190.2

economic and social context (the local annual GDP varies from plant to plant) and level of political trust (see Table 1 for detailed case descriptions).

For each nuclear power plant, the author conducted interviews both with representatives of the operating firm and with local government officials. When interviewing firm representatives, the author asked what they had done to communicate with residents to gain local legitimacy for nuclear project construction; when interviewing local government officials, the author asked them to verify what had been done by operating firms to gain legitimacy and to explain to what extent the local government helped local firms implement these instruments and how they evaluated the effectiveness of the legitimacy tools. In total, 12 interviews were conducted between May 2016 and December 2016 and 21 h of interview transcripts were coded and analysed. Detailed interview information is listed in Appendix A.

Content analysis was used as the main analytical method through which the author identified key instruments and categorised them as knowledge-building or trust-building tools.

4. Main findings

The Chinese government does not design its legitimacy instruments following academic discussions on education and political trust, rather, the National Nuclear Safety Agency (NNSA) requirements designs instruments in four categories: knowledge education and propaganda, information disclosure, public participation and public opinion response. As shown in Appendix B, all of the nuclear power plants studied in this paper used these tools, with variations. Moreover, other instruments were also found during interviews. They were listed under the category of 'other tools'. Three major findings are reported in this section.

4.1. Highly institutionalised legitimacy tool design

The first observation is that the Chinese government regards nuclear legitimacy as a very important issue. An authoritarian state like China does not take the legitimacy of its rule for granted; on the contrary, the Chinese government is highly concerned about possible public opposition to nuclear development. This echoes von Soest and Grauvogel (2015)'s finding that non-democratic rulers also needed legitimating claims to ensure their governance. With a strong top-down administration capacity, the Chinese government not only designates central ministries to establish guidance for gaining legitimacy, but also urges local governments and nuclear firms to build localised institutions for implementation.

4.1.1. Legal guidance for nuclear legitimacy at the central level

At the central level, three ministries and bureaus set the legal requirements for nuclear legitimacy and communication. (Table 2 lists these requests in detail.) The National Development and Reform Commission (NDRC) is in charge of the macro-level planning of the nuclear industry and the administrative approval of nuclear power plant construction projects. The NDRC released *Interim Measures for Risk Assessment of Social Stability for Major Fixed Assets Investment Projects* in 2012 and requested that large investment projects, such as the construction of nuclear power plants, investigate the possible social stability risks by surveying the public, evaluating the degree of risk and proposing ways to mitigate each type of risk. The Ministry of Environmental Protection (MEP), in contrast, is concerned with the possible environmental impact of large projects. It executed *The Environmental Impact Assessment Law of the People 's Republic of China* in 2002 and requested that 199 project types evaluate the possible environmental impact as a necessary condition for obtaining administrative approval.[9]

The NNSA is the key administrative body that monitors, regulates and standardises the safe operation of nuclear power plants. The NNSA has proposed three key documents. The first is the *Guidelines for Public Communication on Nuclear Power Projects (Draft version) of 2014*, which serves as the first national attempt to gain nuclear legitimacy via public participation and communication. In this document, NNSA explains four key questions: Who should implement the gaining of nuclear legitimacy and the relevant communication? What should be included in the communication tasks? How many stages should nuclear communication have? What could serve as communication tools? In 2015, the *Guideline for Public Communication of Nuclear Technology Utilisation* was launched. This document not only extended the public communication requirement to all nuclear-related sectors, but also separated the

[9] The most updated list of projects that need to conduct environmental impact assessments can be found at http://www.360doc.com/content/16/0425/15/9851038_553672743.shtml (in Chinese).

Table 2
Central regulations about nuclear communication in China.
Source: NNSA, 2014, 2015a, 2015b; MEP, 2002; NDRC, 2012.

Regulation Name	Launch Time	Organization	Main Content
Guidelines For Public Communication On Nuclear Power Projects (Draft)	2014	NNSA	• Policy implementers: provincial government, local government, nuclear firms and nuclear developers • Nuclear communication content: public education, participation, information disclosure as well as response to public opinion • Communication stages: in siting, construction, alignment, operation, and decommission stages • Subjects: public who live within 30 km from the nuclear reactor • Knowledge communication tools: publicity lecture, brochure, site visit, special activities, publicity both on traditional media and social media, information disclosure, press meeting, etc. • Participation mechanisms: public opinion survey, public meeting, people's delegation standing committee discussion, feedback to policy, public opinion monitoring and response etc.
Guideline for Public Communication of Nuclear Technology Utilization	Oct. 2015	NNSA	• Extended public communication to all nuclear technology fields including nuclear usage in medical treatment, industry, agriculture, geological exploration, scientific research and teaching • Define public communication in normal situation and emergency situation
Actin Plan On Nuclear And Radiation Safety Public Communication	Nov. 2015	NNSA	• Policy implementers: NNSA, local governments, nuclear firms • Policy goal: institutionalise public participation into nuclear decisions; set up standardized participation mechanism • Publicity format: nuclear knowledge education and propaganda, information publicity, public participation, and public opinion response • Further legal action: add public communication requirements into the draft of nuclear safety law
Interim Measures for Risk Assessment of Social Stability for Major Fixed Assets Investment Projects	2012	NDRC	• For all the large scale projects monitored by NDRC or the State Council, should investigate possible social stability risks by surveying the public, evaluate the risk degree and propose potential solutions to each type of risk • Social stability risks are categorised into three type: 1) the social stability risk is defined as high if majority of the public oppose to the project with strong reactions, or the construction of the project could raise up social mass incidents; 2) the risk is defined as medium if parts of the public show opposition to the project with strong reactions that might lead to social conflicts; and 3) the risk is defined as low if majority of the public ensure the legitimacy of the project, and the oppositions could be released through communications.
The Environmental Impact Assessment Law of the People 's Republic of China	2002	MEP	• Public participation: the State encourages relevant organizations, experts and the general public to participate in EIA in an appropriate way • Public participation format: public hearing or other approaches should be designed to collect the opinions of relevant organizations, experts and the public before submitting the EIA proposal. Project application entity should consider public opinions carefully, and provide explanation as to how they responded to the public opinions.

definitions of public communication in normal situations and in emergency situations. In 2015, the NNSA also launched the *Action Plan on Nuclear and Radiation Safety Public Communication* as the detailed implementation guidelines for nuclear communication. This document required the NNSA, local governments and all nuclear firms in China to institutionalise public participation in the nuclear decision process and called for standard participation timing, format and content. It suggested the following policy tools: education and propaganda, information disclosure, and public opinion response (NNSA 2015). These regulations set up the institutional framework for Chinese nuclear legitimizing and communication.

4.1.2. Localisation and implementation of central guidance

At the local level, the firms that operate each nuclear plant are the main entities in charge of local nuclear communication and legitimacy-gaining. In each plant, a special department is founded in each operating form as the institutional setting for communication. Table 3 lists their names and organisational duties. All of them have two job descriptions: they need to educate local residents in nuclear knowledge and at the same time monitor local public opinion about the nuclear plant so as to ensure the smooth operation of the nuclear power plant via legitimacy-building.

This observation showed that nuclear legitimacy was institutionalised through administrative enforcement and hierarchical governance in China. In an authoritarian country, this ensures a unified national goal for creating nuclear legitimacy. The situation that occurred in the Netherlands, with diverse interpretations of transition goals and managerial tools, did not happen in China, at

Table 3
Organizational setting for nuclear communication departments.
Source: collected from interview information.

	Department name	Duty description on nuclear communication
Qinshan Nuclear Power Plant	Center for press and publicity (新闻宣传中心)	Responsible for the implementation of publicity and media connections
Daya Bay Nuclear Power Plant	Center for art and propaganda under Department of Party and the Public Relationship (党群工作部下属文宣中心)	Nuclear knowledge education within and outside the firm; Community connection activities (i.e. performance and show); Public perception collection; Public communication
Taohuajiang Nuclear Power Plant	Office of Political Work (政治工作办公室)	Nuclear knowledge education; Community connection activities (i.e. performance and show)
Pengze Nuclear Power Plant	Department of Party and the Public Relationship(党群工作部)	Nuclear knowledge education; Community connection activities (i.e. performance and show)

least in the nuclear energy case. Yet nuclear power plants encounter various local situations that require different forms of localisation and policy tool implementation.

4.2. A full range of nuclear legitimacy tool design

China uses a wide range of legitimacy tools in its nuclear energy program with a negligible degree of diversity among nuclear power plants. Unified legitimacy tool design is relatively easy to conduct in an authoritarian system. The NNSA organises regular workshops on communication efforts and offers training programs. Informal learning also happens autonomously and the knowledge is transferred among nuclear firms. This is partly due to frequent human capital transfers among companies and partly due to opportunities such as academic conferences and social media connections. Therefore, if one firm designs a new legitimacy instrument, it soon spreads to other firms. For example, the Daya Bay nuclear company was the first firm to edit a nuclear energy textbook for middle-school students. By 2015, this educational tool had spread to 100 schools in nine provinces with more than 150,000 students involved.[10]

4.2.1. A full range of education tools without clear effectiveness

Following the underlying assumption that more knowledge leads to rational decisions, education tools have been used worldwide as communication instruments for nuclear legitimacy purposes. In the case of China, nuclear knowledge publicity tools include nuclear knowledge posters and brochures; exhibitions; nuclear courses in schools; training programs for governmental officials, community opinion leaders and other members of the general public; nuclear power plant open days; media advertisements and more.

All of the sampled nuclear power plants use knowledge education tools. CGNPC, for example, built nine scientific exhibitions in different nuclear plant sites that hosted more than 250,000 visitors in 2015.[11]

- Nuclear courses in school are regarded as essential for long-term nuclear legitimacy

Both Daya Bay and Qinshan nuclear power plants organise regular campus nuclear education courses in the local community. Taohuajiang and Pengze, although their construction is still pending, also provide nuclear energy courses to local elementary and middle schools on a frequent basis. 'A child can influence a whole family (*regarding nuclear legitimacy*). Offering nuclear courses in elementary school and middle school is quite popular (*in Daya Bay*). Parents who work in the *(Daya Bay)* nuclear station often volunteer to give special lectures to their kids' classes. The company also collaborates with schools to hold the "Light of Charm" nuclear knowledge competition' (Interviews #2, 3). According to the company, more than 200,000 children attended the competition in 2015.

The purpose of campus nuclear courses is not only to provide nuclear knowledge, but also to build trust. 'They (*The children with nuclear education*) don't regard nuclear energy as something awful or dangerous. Rather, they regard it as something normal in daily life, something they are familiar with. Their decisions (*on nuclear legitimacy*) will be more rational than those of people with no nuclear education' (Interview #7).

- Site visiting is regarded as the most effective legitimacy tool

After nuclear programs on school campuses, site visits to nuclear stations are the second favourite tool, according to interviewees. Site visits can be either individual visits to nuclear power plants or organised group visits. The Daya Bay and Qinshan nuclear stations are often cited as excellent examples of 'how nuclear power plants can be technologically safe and economically successful'.[12] All of

[10] Interviews #2, 5, 6, 10.
[11] Interview #2.
[12] Interview #2.

the sites, however, host large numbers of visitors each year. Daya Bay, for example, sets 7 August as its Plant Open Day and invites the public to come for free site visits. In 2016, it hosted more than 500,000 visits.[13]

Nuclear stations in the planning phase or under construction are not allowed to host visitors onsite due to safety concerns. Instead, they usually send local leaders, opinion leaders and other citizens to operating plants such as Daya Bay or Qinshan. This is what the Taohuajiang nuclear plant did before the construction stage to persuade local residents that the technology was safe:

> Villagers, including some village leaders who opposed the nuclear project at first, who feared that local agriculture might be harmed by radioactivity, were invited to visit the Qinshan nuclear plant. They visited the (*nuclear*) exhibition to get basic knowledge and they saw local people living next to the wall of the nuclear station. After they learned that local farmers exported lettuces to Korea with a good profit, they felt that nuclear technology was safe and they became totally supportive. Site visiting is more useful than any other educational tools.
>
> – Interview #7

In China, it is hard to categorise site visiting exclusively as an education tool. Site visiting, according to interviewees, is rather a method of informing the public about how the safety of nuclear technology and how much it can contribute to local economic development and social welfare improvement. In other words, site visiting can significantly increase nuclear legitimacy without increasing citizens' knowledge levels.[14]

- Uncertainties in knowledge educational tools

None of the interviewees denied the effectiveness of knowledge educational tools. 'The nuclear exhibition serves as the starting point. It introduces the basic principles of nuclear reactor operation. It also provides information about nuclear safety, which normally is of most concern to the general public' (Interview #7). 'And the public might feel that the nuclear station is more legitimate if you tell them that the neonatal malformation rate in Qinshan is even lower than in Jiaxing, a larger city farther away from the Qinshan nuclear power plant' (Interview #12). The Qinshan nuclear station once surveyed 50 visitors to the nuclear knowledge exhibition and found that 46 of them increased their nuclear acceptance level after the visit, while the other four kept the same acceptance level. All of these efforts, however, seem to have uncertainties in their efforts.

Although nuclear firms and local governments spend a large share of their nuclear legitimacy budget on educational tools in the hope that citizens will make "rational decisions"[15] (Interviews #2, #10), there is no guarantee that educated citizens will deem nuclear energy legitimate. Even in academic research, the causal link between educational tools and legitimacy levels is contested. Some found a significant linear relationship between knowledge levels and nuclear acceptance (Sjöberg 2001), while others found an inverted U-shaped relationship (Visschers and Wallquist 2013). The same perplexity also exists in Chinese practice: 'We don't really know the outcome of knowledge education. (*We wonder*): will the general public think the project is legitimate if we educate them about how the nuclear reactor works? What if they oppose nuclear energy when they know more about it?'[16]

This puzzle has led to different educational efforts in different nuclear power plants. Some firms emphasise nuclear education and spend a large amount of their budgets on putting up nuclear advertisement boards in communities, painting nuclear advertisement walls, building long corridors of nuclear publicity in cities, designing cartoon figure representing nuclear technology and even shooting movies or cartoons for nuclear knowledge education.[17] Other firms do not believe in the importance of education communication and consequently have limited budgets for nuclear education.

In addition, it usually takes a long time for educational tools to show their effects. Therefore, local governments and nuclear firms usually use educational tools for daily communication, but select other tools (such as economic stimulation) to cope with urgent legitimacy crises.

4.2.2. Selective public information disclosure tools

In each nuclear power plant, the organisation in charge of legitimacy-gaining also takes charge of nuclear information publicity. The content of the publicity varies from plant to plant. Daya Bay, for example, has a regular publicity mechanism and reports all levels of nuclear and radiological events to the media and the public. The other three nuclear power plants have WeChat[18] accounts and publish relevant news and notices via this social media tool. They do not regularly convey information directly to the media.

Information disclosure also requires frequent educational efforts, directed both to the media and to the public. Daya Bay, for example, encountered a recent round of public misunderstanding among Hong Kong residents:

> The company decided to release information on nuclear and radiological events starting in 2013. According to the International Atomic Energy Agency (IAEA), nuclear events are scaled into 12 levels, with a level 0 event being the least severe.[19] In 2013 Daya Bay reported several level 0 events that brought no threat to the safe operation of the station. However, many media reports

[13] Interview #2.

[14] A detailed discussion of legitimacy increases is found in 4.3.3.

[15] "Rational decision" is the term some interviewee's adopted very often to describe the logic that citizens would have higher level of nuclear legitimacy and acceptance when they have more knowledge of nuclear technology.

[16] nterview #10.

[17] Interviews #2, 10.

[18] WeChat is a Chinese social media application.

[19] For example, leakage from the primary cooling circuit is a level 0 event. It is also a level 0 event if the operator in the control room forgets to close the door.

published this news with the headline 'Daya Bay had nuclear leakage accidents in 2013′, which generated severe panic in Hong Kong and other areas close to the plant. We had to call all of the media outlets using the inaccurate headline and educate them on the difference between a level 0 event and radioactivity leakage. And we asked them to either withdraw the report or to correct their mistake… That was a ridiculous mistake, but it happens a lot in public information disclosure.

– Interview #2

Therefore, some plants regard information disclosure as risky and are reluctant to actively report detailed information about operational data, safety status and so on.

4.2.3. *'To cope with the errand': reluctant implementation of participation tools*

In all of the regulatory documents released by the NNSA, MEP and NDRC, public participation is repeatedly emphasised as a necessary step in nuclear project construction. The NNSA states that 'there should be clear design of public participation, including in the participation stage, procedure and organisational format. There should also be an official monitoring mechanism to check the implementation of public participation' (NNSA 2015a). However, the interviews revealed a limited and reluctant implementation of this legitimacy tool. Daya Bay invited Hong Kong residents to Beijing once in 1986 to express their worries about the project and participate in the deliberation process. Taohuajiang and Pengze conducted public opinion surveys for their environmental impact assessment report and social stability risk impact assessment report, respectively. No additional participation instruments were found in this study. In some cases, interviewees pointed out the vagueness and conflict design in participation tools.

There is no doubt that we regard public interest and public participation as important elements in nuclear development. However, we are confused by different administrative requirements. At the early stage of nuclear plant construction, we have to do the same thing (*i.e., public opinion surveys and public participation meetings*) three times at the same time: one to (*get approval for*) the environmental impact assessment (*required by MEP*), one for the social stability risk impact assessment (*required by the NDRC*) and one for the public communication requirement (*set by the NNSA*).

– Interviews #4, 6

The interviewees indicated that they regarded public communication and maintaining nuclear legitimacy as lifelong tasks that start with nuclear station construction and end after the station is decommissioned. Yet they did not have time and energy to design communication plans to meet this long-term goal. They had to fulfil short-term administrative requirements first. In other words, they needed to report three different public opinion surveys to three different bureaus for administrative approval. To make life easier, they simply combined three surveys into one, although the three surveys were supposed to cover different aspects of public opinion on nuclear legitimacy.

This segmentation at the central level reflects a typical transition paradox. Transition challenges are usually innovative and comprehensive in nature and require collaboration, experimentation and learning on the part of transition management (Voß and Bornemann 2011). A traditional governance system with separated administrative functions is unlikely to handle transition tasks well. In the Chinese system, fragmented central authorities create disjointed policymaking because policy initiatives 'need to acquire the active cooperation of many bureaucratic units that are themselves nested in distinct chains of authority' (Lieberthal and Oksenberg 1988, p. 22). Nuclear legitimacy-gaining, in this sense, is not an exception.

Second, the purpose of maintaining transition legitimacy according to the Western literature is to ensure a channel through which the general public can express its interests so as to increase public acceptance of policy initiatives (Hendriks 2009). However, Chinese local governments and nuclear firms are so used to interregional and interorganisational competition that they try to make the public acceptance rate as high as possible, although no official bar has ever been set. In this sense, they do not genuinely care to learn about public opposition and find solutions in response, but regard high survey scores as their ultimate goal.

Two-way participation tools, such as the status meetings held in Western countries, were not observed in this research. The only exception was the Daya Bay case in 1986, when Hong Kong citizens showed strong opposition to the first phase of the project due to fears generated by the Chernobyl nuclear accident, which had happened earlier the same year. By that time, Hong Kong was still a colony of the UK, and the Chinese central authority had no direct administrative or political control over the area. Therefore, the opposition was regarded as both a policy legitimacy issue and a political issue. 'We (*Daya Bay Nuclear Corporation*) used all of the methods we could think of to gain legitimacy and acceptance (*among the local residents*). On the one hand, we collaborated with the Hong Kong local media, such as newspapers, television stations and subways, for intensive public education. On the other hand, NGOs and opinion leaders who opposed the project were invited to Beijing to express their views directly to central leaders.'[20]

After Hong Kong's handover, further public opposition in Hong Kong was handled by Daya Bay rather than by the central government. One-way information disclosure is more often used than two-way communication tools. Prudence in the use of participation tools in China also reflects Wynne's (2010) argument that appreciation of technology is the prerequisite rather than the result of opening up policymaking processes. In other words, if there is insufficient acceptance of nuclear energy, opening up the process of nuclear decision-making may not be successful, but may instead cause administrative challenges.

4.2.4. *Collaborative public opinion monitoring tools*

Social stability is a high priority both for local governments and for nuclear energy plants. To closely monitor public opinion

[20] Interview #3.

about a plant, special groups are usually set up within the nuclear power plant and in local government to continuously monitor all of the relevant media reports, online comments and discussions and breaking news. 'If any negative comments or opinions appear, we need to either contact the media or find the author for direct communication. The ultimate purpose is to withdraw the comments and to reduce the level of opposition' (Interviews #1, 11).

Monitoring efforts require close collaboration between local government and the nuclear power plant. Sometimes, however, the local government has other priorities. Local officials' promotions are vetoed if there is social instability (e.g., protests, riots or petitions) within the territory. Therefore, some local officials prefer not to use participation tools out of fear that they may draw public attention to the power plant, which may lead to social instability (Interviews #4, 5, 6).

4.3. New communication tools to gain nuclear legitimacy

During the data collection process, this research also found several legitimacy tools that were widely used in China but rarely reported in the literature. The purposes of these tools were simply to 'increase the local legitimacy of the nuclear project'.[21] This goal echoes Fang's (2014) work on the psychological substitution of technology judgments. In his research, Fang observed that local people replaced their trust in government with trust in nuclear technology. Along the same line, policy tools were designed to increase nuclear legitimacy by connecting nuclear energy to many aspects of people's daily lives.

4.3.1. Integrating nuclear energy into local culture and social norms

All of the interviewees expressed the belief that local people would accept nuclear technology if they grew used to it in daily life. Consistent with this belief, many policy tools were designed (as shown in Appendix B under 'others'), such as using local folk art to introduce nuclear knowledge, organising local opera performances with nuclear advertisements during the intermission, donating titled scholarships to schools and setting up special funds to help widows, aging people, orphans and poor people in the local community.[22] In each hotel in Taojiang County, for example, guests can find nuclear brochures on their nightstands, and all local tourist advertisements mention the nuclear power plant as one of the local attractions. With all of these legitimacy-oriented tools, nuclear companies hope that the concept of nuclear energy will be hammered into local people's minds. This method is used in particular in the early planning and construction stages of a nuclear project.

4.3.2. Economic benefit-sharing as a local legitimacy-gaining tool

Among all of the legitimacy tools, the effectiveness of economic benefits is one of the most obvious, according to interviewees. Many regarded it as the 'ultimate effective method' to persuade local people and the local government to accept nuclear technology. Villagers can benefit from nuclear station construction from its beginning. The Pengze project relocated 1500 villagers to a nearby county where immigrants not only lived in new townhouses at a very low price, but also could send their children to county elementary schools to receive a better education. Villagers who ran restaurants and hotels near the construction site earned much more than before.[23] These direct benefits served as significant and immediate stimulations of local acceptance.

Potential long-term economic benefits are also persuasive evidence both for local governments and local residents.

> If the Qinshan mountain (the hill where the Qinshan nuclear reactors locate now) were not used for nuclear industry development, it would have still been a wasteland without any economic income. But right now, it (*the nuclear company located on Qinshan mountain*) generates 20 billion yuan in taxation each year. I persuaded Taohuajiang county leaders (*to accept nuclear energy in their hometown*) using this simple fact; it was very effective.
>
> – Interview #11

Today, economic development still serves as an important evaluation item at the local level. It is therefore not surprising that low-income regions are more likely to legitimize nuclear projects. One interviewee from Taohuajiang even said that local residents were so eager to have the nuclear project that they would have petitioned to keep it if the project had continued to have a 'pending' status.

The literature examines how economic benefits might facilitate nuclear development at the local level. However, the causal relationship has not always been significant and positive (Flynn et al., 1992) and its influential mechanism varies in different situations (Gregory and Mendelsohn 1993). This paper found that the effectiveness of economic stimulation tools might decrease as the local economy grows. Haiyan county, where the Qinshan nuclear station is located, now has a high level of per-capita GDP and local residents and government have become reluctant to allow new nuclear reactors. Instead, they prefer manufacturing-based nuclear projects that can contribute to local economic development at a much lower risk. This complicated balance between economic development desire and risk aversion makes it hard to predict in what situations nuclear projects may encounter legitimacy challenges in China.

4.3.3. Issues with legitimacy-oriented tool design and implementation

In practice, legitimacy tools help nuclear firms gain public acceptance during the project construction process. However, their long-term effects are still not evident to many nuclear plants:

[21] Mentioned in interviews # 2, 5, 6, 7, 10, 12, 13.
[22] Community activities are conducted in all cases except for Pengze, according to interviews #2, 3, 7,10, 11.
[23] Interviews #4, 5, 6.

> In my opinion, nuclear communication and legitimacy maintenance should last for at least 100 years, which includes siting, design, construction, operation and decommission. Economic stimulation only serves (*as a useful legitimacy tool*) at certain stages. Citizens will consider other issues during this time and we (*the local government*) need to develop new communication tools in response to these changing opinions.
>
> –Interviews #11, 12

The legitimacy tools mentioned in this research are designed to showcase nuclear benefits or to encourage people to take the existence of nuclear stations for granted. Citizens can be 'brainwashed' by intensive legitimacy tools; however, the literature shows that this effect may be temporary and can be easily altered by other risk events in the long run (Parkhill et al., 2010). Interviews also found that citizens would be highly likely to attribute local environmental degradation to nuclear energy no matter how long the plant had existed locally.[24] In such cases, the legitimacy tools mentioned in these four cases could hardly solve legitimacy challenges in the long run.

In the Western literature, various democratic formats have been tested as ways to gain transition legitimacy. Deliberation, inclusion, transparency and openness are basic principles for better transition management (Hendriks, 2009; Jhagroe and Loorbach 2015; Stirling 2008). In China, concurrent legitimacy tools, although effective in the short run, may still need to be combined with other participation tools to gain long-term effectiveness.

5. Conclusions and implications

This paper focuses on the nuclear legitimacy tools used by Chinese nuclear firms and local governments in the past 30 years. Along with increasing concern for environmental quality, the general public in China has started to express its oppositions on nuclear development. This development requires local government and nuclear firms to design suitable policy tools to gain legitimacy for nuclear power. Yet, issues around public participation in energy transitions in non-Western democratic contexts have been an under-researched area. To fill this gap in the literature, this paper uses four nuclear power plants as sample cases and finds that the Chinese government has systematically institutionalised nuclear legitimacy tasks with umbrella regulation guidance at the central level and reflexive and detailed policy instrument design at the local level. Nuclear firms and local governments collaboratively use education-oriented instruments, information publicity tools, participation tools, public opinion monitoring tools and other legitimacy-oriented tools to gain nuclear legitimacy. Each type of policy tool is conditionally effective. Combined policy tool design and long-term policy tool design is recommended.

The literature has proposed that a strong centralised authority might provide a better institutional setting to meet long-term goals such as green development or sustainability transitions (Heilbroner 1974). This paper finds advantages in a centralised governance structure in the Chinese nuclear legitimacy case. Nation-wide regulations and institutional settings are easier to standardise, and the implementation of central policy tools is smoother with centralised administrative evaluation and political promotion assessment.

Meanwhile, this paper also reports several disadvantages of a centralised authority in meeting long-term transition challenges. First, it is still hard to design effective long-term tools due to fragmented interests. In a democratic system, plural interests bring political considerations into transition management, which normally leads to activities such as reframing and renegotiating the meaning of legitimacy (Avelino et al., 2016). A centralised governance structure, however, has other types of fragmentation. Competition among administrative bureaucracies, the terms of political nomination and assessment and conflicting policy goals set by different ministries are sources of fragmentation that can prevent centralised structures from achieving sustainability transition tasks in the long run.

Second, democratic systems have to redefine the most basic democratic concept, 'the public', to explain other principles such as inclusion, deliberation and representativeness (Hendriks 2008, 2009). The authoritarian system in China also encounters the need to redefine 'the public'. The key questions are: Who are the public? What are the public's interests? For now, Chinese nuclear firms take local residents as 'the public' during the project preparation stage and take people with opposing opinions as 'the public' during the operational stage. This identification bias explains why the concurrent communication process may be quite successful at the preparatory stage yet may encounter many challenges during the operational stage: ignoring possible interest groups from other places makes nuclear firms passive. They may hardly notice the opposition unless it bursts out.

Third, even authoritarian governance has to consider the cost and effectiveness of legitimacy tools, particularly evaluating the relationship between the communication scale and the cost. In practice, nuclear firms set up selective communication plans that communicate only with immigrants during the immigration and construction period, with local villagers and residents during the operational period and only with government officers for administrative assistance. As the interviewees indicate: 'MEP requires full public support within a 5-km diameter around the nuclear site to pass the environmental impact assessment review. No company really has the motivation to conduct a public opinion survey beyond 5 km, not to mention 30 km (as required by NNSA). … Public opposition sometimes comes from other places beyond 30 km distance. The nuclear plants would be bankrupt if they covered all of the citizens who might be stakeholders'.[25] As far as the author can tell, no theoretical guidance exists to help Chinese nuclear firms optimise their communication cost-benefit ratio. Which communication tools are most effective? How should they balance the ratio between knowledge-based tools and political trust tools? What tools should be adopted, modified or eliminated? All of these

[24] Interviews #2, 10, 11.
[25] Interviews #10, 4, 6.

questions need further exploration.

Furthermore, there is still a need for additional theoretical and empirical guidance on the appropriate degree to which nuclear decisions should be opened to the public (Lehtonen 2010; Stirling 2008). Democratic countries, which are 'open', have encountered innumerable administrative and political delays and have been puzzled about what to open, how to open and what the results of opening might be. Meanwhile, China, which is highly centralised and should not have to be opened up, faces challenges from an increasing amount of public opposition and social instability, which has gradually forced China to open its nuclear policymaking process. The purpose of the opening, however, is different from Western democratic participation that brings public opinion into decision making process. Rather, nuclear decision openness in China is more of a tool to increase nuclear legitimacy so that oppositions will cease. The participation opportunity is even designed downstream from nuclear firms or the government, which reflects its administrative function rather than democratic function. In the long run, China, too, has to face the questions of what to open, how to open and what the results might be.

Legitimacy-gaining tools are essential to the design and implementation of sustainability transition goals. To fulfil its rapid nuclear development strategy, the Chinese nuclear industry needs to check the effectiveness of its concurrent legitimacy-gaining instruments to ensure the sustainable development of the technology.

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Appendix A. Interview information.

	Department	Interview Length (hours)	Interview Date
NNSA	Department of nuclear safety	2	May, 2016
Daya Bay	Center for art and propaganda	2.5	August, 2016
	Administrative office	1	July, 2016
Pengze	Jiangxi Nuclear Corporation	1	Nov. 2016
	Jiangxi Nuclear Corporation, Department of Party and the Public Relationship	2	Nov. 2016
	Pengze county	1	Nov. 2016
Taohuajiang	Taohuajiang Nuclear Corporation, Office of Political Work	1.5	Nov. 2016
	Taohuajiang Nuclear Corporation	1	Nov. 2016
	Taojiang county	1	Nov. 2016
Qinshan	Qinshan Nuclear Corporation, the Center for press and publicity	1.5	July, 2016
	Haiyan County, the Office of nuclear supporting and emergency	2.5	July, 2016
	Haiyan County, the Office of nuclear supporting and emergency	1	Dec. 2016
	Qinshan Nuclear Corporation, Department technological support	3	July, 2016

Appendix B. Detail Information about Communication Tools Adopted during Chinese Nuclear Development

	Instrument Type	Qin Shan	Daya Bay	Taohuajiang	Pengze
Knowledge education and propaganda	Nuclear knowledge education	• Build national nuclear knowledge exhibition hall for visitors • Build up a nuclear education network among local government, nuclear firm, city party college, city science and technology	• Nuclear knowledge brochure for adults and for kids • Special educational program to Hong Kong residents via media and exhibition • Set formal course of nuclear topic in elementary schools, and in	• Scientific exhibition in Yiyang city and Taojiang county with more than 12,000 local visits • Design official mascot for the nuclear station and broadcast it on local TV	• Nuclear educational training programs in government sectors, community, and schools • Set nuclear knowledge column in local newspaper • Organizing nuclear knowledge contest

		association, city bureau of education, and city bureau of public health • Set up training to nuclear experts, teachers, doctors, village leaders twice a year and make this group as essential members of nuclear education • Set up local media network that includes online and offline medias to broadcast advertisement • Propaganda nuclear knowledge in government sectors, in schools, and in community	universities, organising summer camp in nuclear plants • Regular media press publicity	• Cast TV program with Hunan Satellite TV to publicize knowledge about nuclear technology and proper emergency response	in schools and in communities • Inviting nuclear experts to give speech in communities
	Educate potential environmental benefit	Introducing Qin Shan as one of the countries with best air quality where many people lived over 100 years old	N/A	• Educating people that nuclear development could reduce smog	N/A
	Site visiting	• Set up schedule 6–8 open days per month and target 100 thousands visits within three years	• Annual open day to the general public for visiting • Organizing site visit in government sectors, schools, media reporters on regular bases	• Inviting local people to visit Qinshan nuclear station • Site visits were well organised. For example, it strictly follows the schedule and prepares customized dinner menu for visitors to show the organisational capacity of the nuclear plant • Local officials and opinion leaders post their visiting itinerary online for sharing	• Organize leaders of the village to visit Qinshan and Daya Bay nuclear station for its advance in economic development with safety
Information publicity		N/A	• Disclosure all levels of nuclear and radiological events to the media starting from 2013	• Maintaining Wechat account as official publicity site	• Maintaining Wechat account as official publicity site
Public participation		N/A	• Invited citizens with oppositions of Daya Bay to Beijing to express their	• Conduct public survey both in Environmental Impact Assessment	• Conduct public survey both in Environmental Impact

			views directly to central government leaders in 1986	stage and in Social Stability Risk Assessment stage among the 5 km diameter range of the site	Assessment stage and in Social Stability Risk Assessment stage among the 5 km diameter range of the site
Public opinion response	Public opinion monitoring		• Check online posts and news for anything related to Daya bay nuclear station or with nuclear technology. Clarify rumors and correct wrong information and reports	• Form public opinion and communication group jointly led by the provincial leader, local officials and the office of political issues in the nuclear firm	
	Bonding with local government	• Build-up close relationship with local government to facilitate nuclear project authorization • Limits public communication within Haiyan county, especially in several villages next to the station	• Collaborate with local government and local NGOs in Hongkong to do nuclear education		• Organizing training programs especially for party members and local officers
Others	Environmental benefit	• Introducing Qin Shan as one of the countries with best air quality where many people lived over 100 years old		• Educating people that nuclear development could reduce smog	
	Building trust to the company	• Qinshan nuclear project started as a military project named “728 project”, which represented for national security interests • Safe operation for 25 years is the best evidence of management capacity	• Learn from French government about nuclear technology and communication based on technological transfer relationship with France	• State-owned company has to consider about public interests. They will consider water shortage for sure, unlike private companies.	
	Collaborative community activities	• Organizing celebration party for the 20th years of Qinshan II project. Inviting local residents and local workers to joint the celebration • Collaborate with local schools and host their graduation	• Helping local communities to build “friendly, warmly, and safe community” • Organizing “Nuclear Culture Performance Party” that promotes nuclear development using folk art format such as paper	• Organize local opera performance with nuclear advertisement • Set up special funds to help widows, aging people, orphans and poor people • Staffs and technological experts of nuclear station live close to	• Nuclear knowledge publicity during folk activities • Set up special funds to help widows, aging people, orphans and poor people

	ceremonies in nuclear station. Set special scholarship in local schools	cutting, kite flying, hydrangea dance etc.	nuclear station to show environmental safety	
Participation	• Communicate with residents who has different oppositions in person	· Invite public and NGOs in Hongkong with different opinions to talk directly with central government		Communicate with residents who has different oppositions in person
Economic benefit to local development	• Local farm products exported to Korea to show how safe the nuclear could be • Hire more than 10,000 local residents to work in nuclear station • Spend 0.5 billion RMB for local highway construction • Plan to build "nuclear industry zone" in local city as nuclear industry spill effect. It is estimated that the annual production worth 20 billion RMB	• Helping local residents sale special products, such as seafood, cherry, tea and others.	• Pay for the construction of local infrastructures such as tunnels and roads, which greatly shorten the time locals travel from the village to the county • Estimated annual taxation of 3 billion RMB • Donate 6 billions for local development • Decrease Hunan's reliance on coal fire plant and increase electricity supply, reduce electricity price	• Well designed demolition plan, through which each family has a new townhouse in the town • Plan to share nuclear taxation, human capital, job opportunity, and revenue with nearby province within 5 km distance to boost regional development regardless of administrative boundary • Help Jiangxi to its energy structure and reduce reliance on coal, oil, and hydro power, which need to be imported from other provinces

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